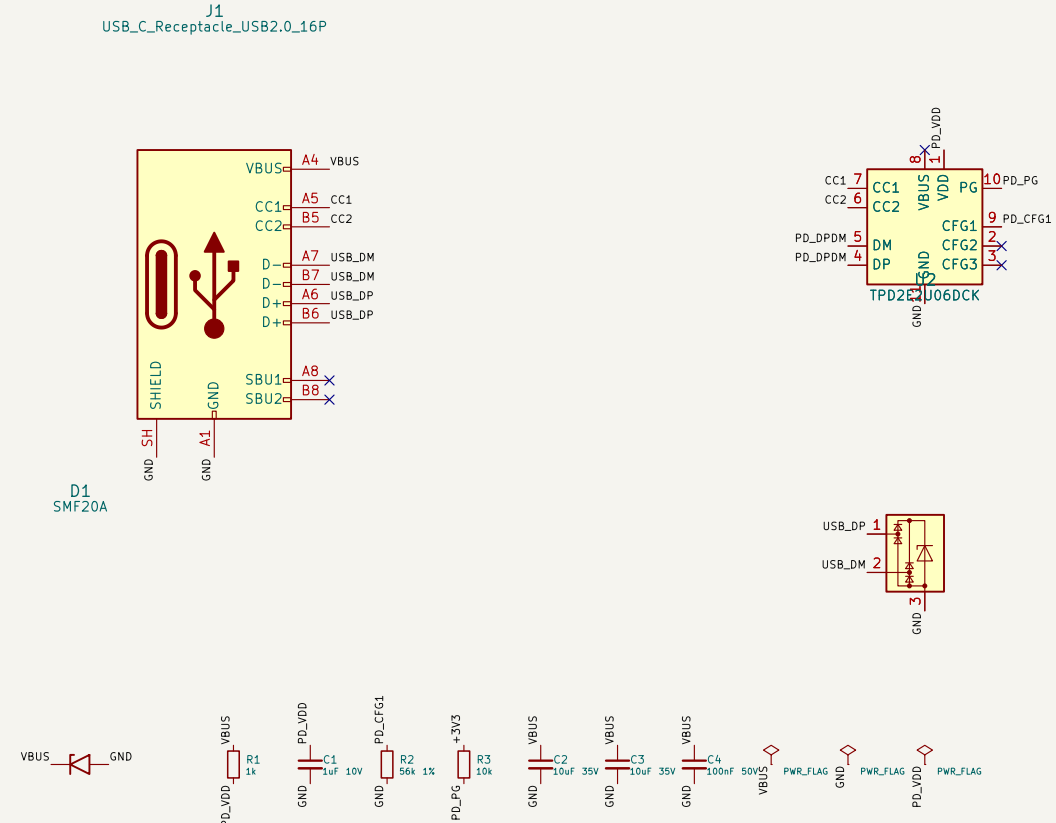


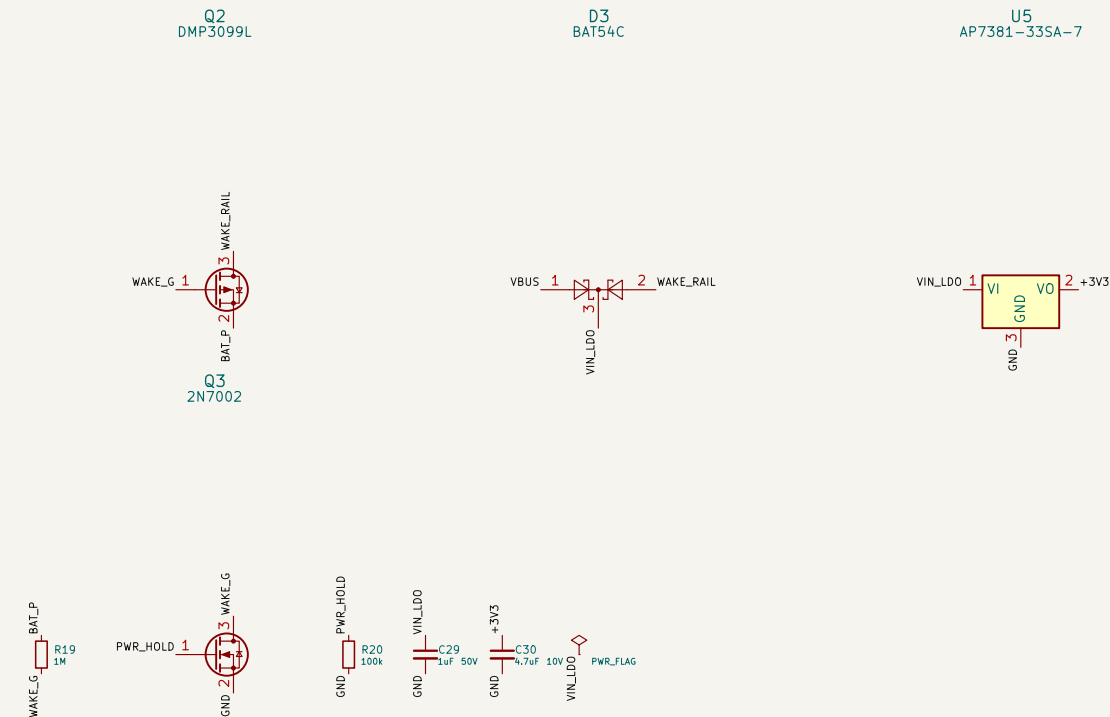
1. USB-C INPUT + PD SINK (requests 15V)

CC1/CC2 direct to CH224K (Internal Rd). DP/DM of CH224K shorted together and NOT connected to the connector (WCH 'PD-only' mode) so USB D+/D- go to the MCU. CFG2/CFG3 float = single-resistor mode. CFG1 56k = 15V. PG low = contract OK. CH224K VBUS sense pin left NC (allowed in PD-only mode; pin is only 13.5V rated).



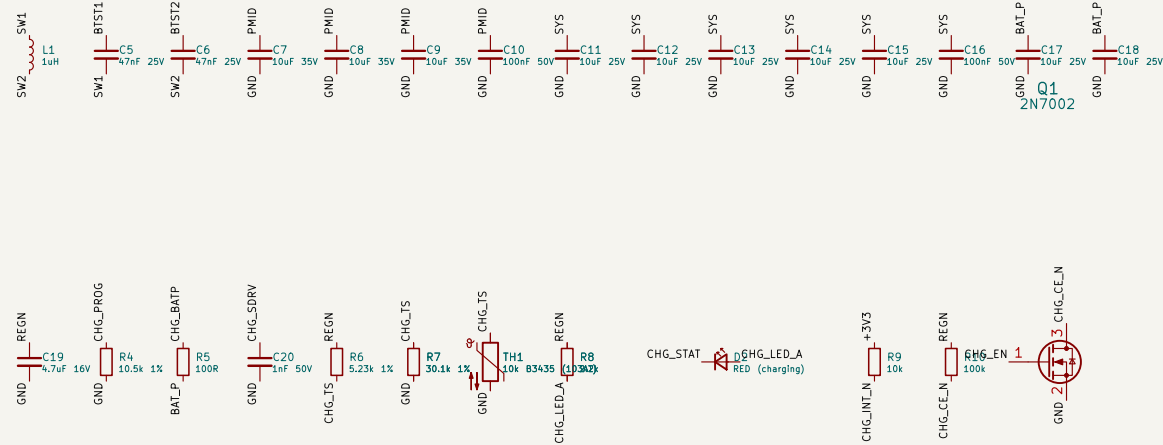
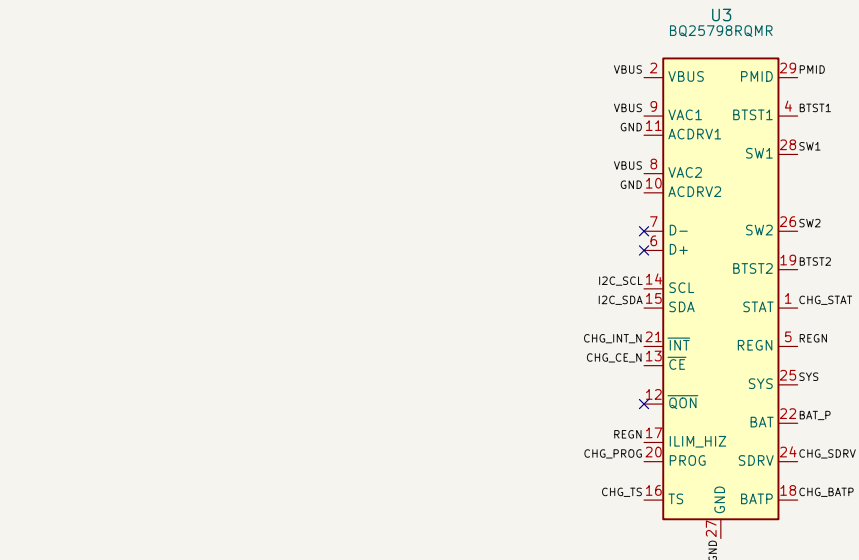
4. MCU POWER (USB / self-hold)

3V3 = OR of: VBUS (charger OR a phone in the USB-C port) | battery via Q2 when the MCU asserts PWR_HOLD (lets the MCU finish balancing, put BQ25798 in shutdown + BQ76920 in SHIP after unplug, then release itself). Nothing else can power the MCU from the pack. Idle battery drain ~1-2uA total.



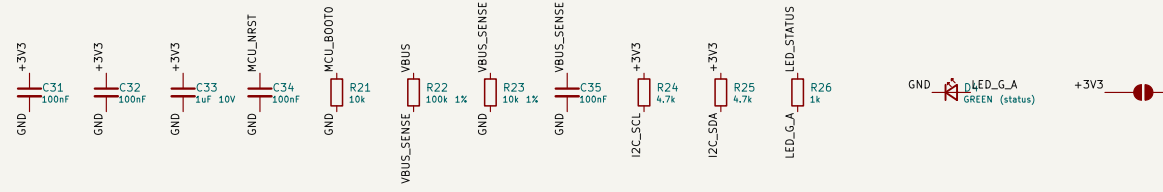
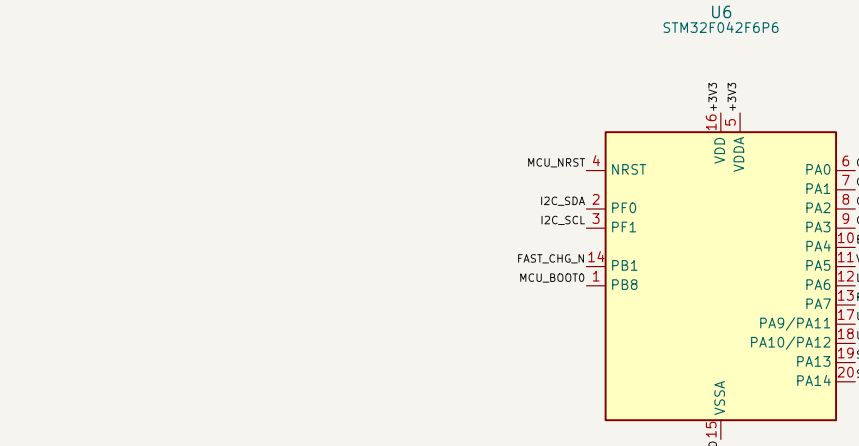
2. 3S CHARGER BQ25798 (buck-boost, 0x6B)

PROG 10.5k -> 3S / 1.5MHz; POR defaults VREG=12.6V, ICHG=1A, VSYSMIN=9V. No ACFTs: VAC1/VAC2 tied to VBUS, ACDRV1/2 to GND, ILIM_HIZ->REGN (limit set by I2C). /CE held HIGH by 100k -> NO charging until MCU drives CHG_EN high (after cell check). BQ25798 D+/D- left open: MCU must clear AUTO_INDET_EN and set IINDPM over I2C. TS: 5.23k/30.1k + 10k NTC = 0.60C window (datasheet 7.3.9.5). Shutdown mode = 0.5uA.



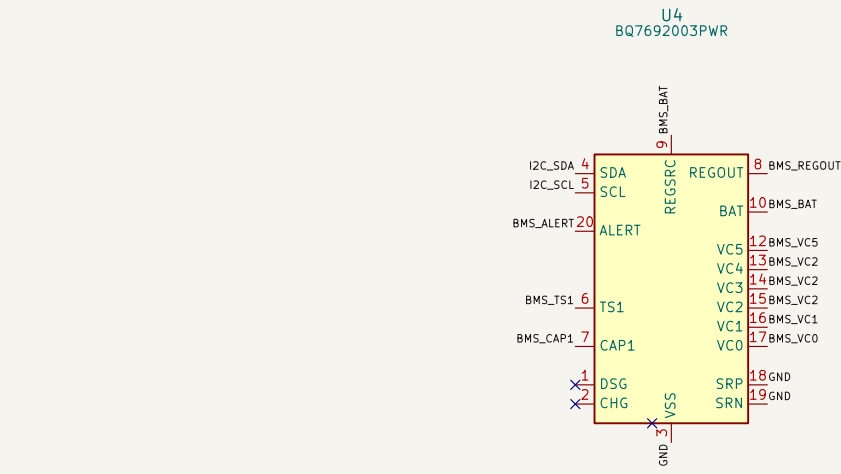
5. MCU STM32F042F6P6 (TSSOP-20, USB FS, DFU)

PF0/PF1 = I2C1 SDA/SCL (AF1). PA11/PA12 remapped onto pins 17/18 for USB (SYSCFG PA11_PA12_RMP). PA1/PA3/PA0 = charlieplexed cell LEDs A/B/C (pads TP13-15). Charger /INT, CH224K PG and BMS_ALERT are not wired to the MCU (firmware polls I2C). PA2 CHG_EN, PA4 BMS boot, PA5 VBUS/I1 ADC, PA6 LED, PA7 PWR_HOLD, PB1 = FAST_CHG_N (JP2 to GND, internal pull-up); open = slow charge (default), bridged = fast charge. Bridge JP1 + plug USB = ST ROM DFU bootloader. Backup: ST-Link on test pads TP5 SWDIO, TP6 SWCLK, TP7 3V3, TP8 GND.

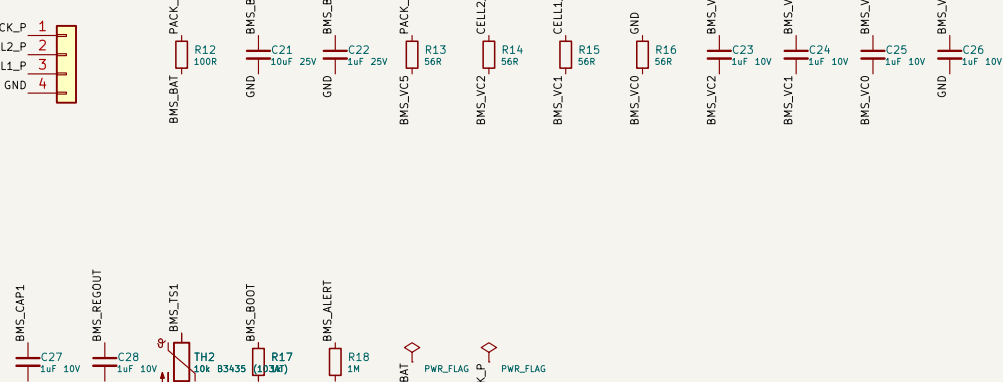


3. CELL MONITOR/BALANCER BQ7692003 (0x08)

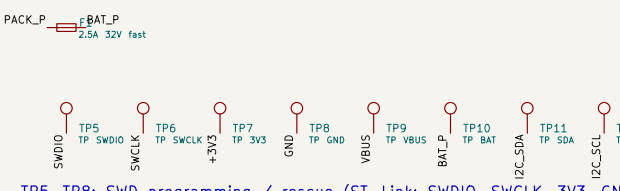
3S per datasheet Table 9-2: VC5-VC4 = cell3, VC4-VC3+VC2 shorted, VC2-VC1 = cell2, VC1-VC0 = cell1. Rc=56k / Cc=1uF cell filters -> ~37mA internal balancing (datasheet min Rc 40R, max 50mA) (never balance adjacent cells together). SRP/SRN to VSS (no coulomb counter), CHG/DSG unused, Sits in SHIP mode (-0.6uA) when idle. Balance lead also carries the charge current (pins 1 and 4): firmware pauses charging before each cell reading.



J2 BALANCE 3S (JST-XH 4P)



6. CHARGE PATH (through the balance plug J2)



TP5-TP8: SWD programming / rescue (ST-Link: SWDIO, SWCLK, 3V3, GND) on the back. TP9-TP12: VBUS, BAT, I2C SDA, I2C SCL probe pads. The pack connects ONLY through the JST-XH balance plug J2 (no Deans / main-lead pads), Charge current: J2 pin 1 (pack +) -> F1 2.5A fast -> BAT_P (charger), return via J2 pin 4 (GND). J2 pin order follows the pack's plug: its black wire lands on header pin 4 (checked on the real pack). F1 protects the thin balance wires: a board short would otherwise burn them inside the stock. Max charge 1.4 A (JP2 bridged). JST-XH is rated 3 A on AWG22; keep 26 AWG wires at <= ~2 A. Field check: plug a phone (USB-C DTG) into the USB-C port -> MCU wakes on the phone's 5V, reports V + % per cell over USB; charges only at 12-17 V.



H1/H3 (USB-C end): M1.6 (1.7 mm) so heads clear the USB shell. H2/H4: M2 (2.2 mm). Concentric with R3 corners.

7. CELL LEDs (3 wire pads near the USB-C charlieplexed)

Remapped LED board: three 2-pin red/green bi-colour LEDs (one per cell) wired to a trianode cell 3 between LED_A-LED_B, cell 2 between LED_B-LED_C, cell 1 between LED_C-LED_A. Red LED anode on the first pad of the pair, green anode on the second pad, both cathodes to GND wire. Red = cell charging, green = cell full: lit only while charging -> mA peak 1/3 duty.